Simulation Modeling and Analysis

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Introduction

- The basic ingredient needed for every method of generating random variates from any distribution or random process is U(0, 1) random variables.
- For this reason it is essential that a statistically reliable U(0, 1) random-number generator be available.
- Without an acceptable random-number generator, it is impossible to generate random variates correctly from any distribution.
- There are usually several alternative algorithms that can be used for generating random variates from a given distribution, and several issues should be considered when choosing which algorithm to use in a particular simulation study.

- The first issue is exactness. We feel that, if possible, one should use an algorithm that results in random variates with exactly the desired distribution, within the unavoidable external limitations of machine accuracy and exactness of the U(0, 1) random-number generator.
- Given that we have a choice, then, of alternative exact algorithms, we would clearly like to use one that is efficient, in terms of both storage space and execution time.
- Some algorithms require storage of a large number of constants or of large tables, which could prove troublesome or at least inconvenient. As for execution time, there are really two factors.
- Obviously, we hope that we can accomplish the generation of each random variate in a small amount of time; this is called the marginal execution time.
- Second, some algorithms have to do some initial computing to specify constants or tables that depend on the particular distribution and parameters; the time required to do this is called the setup time.

- In most simulations, we shall be generating a large number of random variates from a given distribution, so that execution time is likely to be more important than setup time.
- If the parameters of a distribution change often or randomly during the course of the simulation, however, setup time could become an important consideration.
- A somewhat subjective issue in choosing an algorithm in terms of its overall complexity.
- There are many techniques for generating random variates, and the particular algorithm used must, of course, depend on the distribution from which we wish to generate; however, nearly all these techniques can be classified according to their theoretical basis.

Inverse Transform